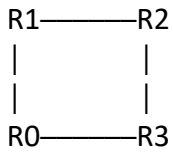
Lab1: Supernetting & Aggregation in a 4-Router Square

Objective:
Build a 4-router square (R0, R1, R2, R3). R1, R2 and R3 each own one LAN subnet. Choose subnets so that:

1. R1 and R2 subnet sizes are **equal**.
2. R3 subnet size = (R1 subnet size) + (R2 subnet size).
3. All three LAN subnets are **contiguous** (placed next to each other in address space).
4. The **first IP (the starting address of the block)** must be aligned (divisible) by the **total** block size covering the three subnets (i.e., must be the natural boundary for the aggregate).
5. Configure routing so **R0 holds a single supernet route** covering all three LANs (i.e., R0's routing table shows the supernet, not three separate routes).



Physical links: R0–R1, R1–R2, R2–R3, R3–R0 (each a point-to-point link).

Tasks (step-by-step)

1. Build the topology in Packet Tracer with four routers (name them R0..R3).
2. Assign LAN subnets for R1, R2, R3 following the constraints above (pick sizes and addresses).
3. Assign /30 point-to-point IPs for the four inter-router links.
4. On **R0** configure a **summary address** that aggregates the three LANs into a single supernet. Verify that R0's routing table shows only the supernet (one entry) instead of three separate networks.
5. Verify connectivity (ping from host in R1 LAN to host in R3 LAN, etc.) and show routing tables.

Use: Cisco 2811 Router

Because:

- It supports **WIC-2T** or **WIC-1T** serial interface modules
- It has **2 built-in FastEthernet ports (Fa0/0 and Fa0/1)**
- After adding **one WIC-2T card**, you get **2 serial ports (S0/0/0 & S0/0/1)**
- Perfect for "square-shape" router-to-router serial connections